

2017 Ph H2 Q16

Section: Electricity

Topic: Current, PD, Power, Resistance

A resistor of resistance $R = 150\ \Omega$ is connected across a power supply of $V = 12\ \text{V}$.
The resistor is used for 2.0 minutes.

(a)

Calculate the current in the resistor.

Use Ohm’s Law:

$$I = \frac{V}{R} = \frac{12}{150} = 0.080\ \text{A}$$

(b)

Calculate the energy transformed in the resistor during this time.

Step 1: Convert time to seconds:

$$t = 2.0\ \text{min} = 120\ \text{s}$$

Step 2: Use:

$$E = IVt = 0.080 \times 12 \times 120 = 115\ \text{J}$$

(c)

Calculate the power dissipated by the resistor.

Use:

$$P = IV = 0.080 \times 12 = 0.96\ \text{W}$$

✅ Alternatively, use:

$$P = \frac{V^2}{R} = \frac{144}{150} = 0.96\ \text{W}$$

🧠 Revision Tips:

- Ohm’s Law: $V = IR$
- Energy: $E = IVt$ or $E = Pt$
- Power:

$$P = IV = I^2R = \frac{V^2}{R}$$

Choose the formula that matches the data you’re given!